



Escaping the vendor trap

A journey for migrating legacy infrastructure to OpenStack and K8s

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EVENT

KubeCon + CloudNativeCon + OpenInfra Summit + PyTorch Conference China 2026

Shanghai · September 2026



Introduce Authors



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Agenda

05 chapters — how Viettel escaped the vendor trap and built open infrastructure ownership

01

Legacy

The pain that forced us to break free

02

Strategy

Choosing where to start

03

Migration Journey

From pilot cluster to platform factory

04

Outcome

80%+ on open architecture — three sovereignties

05

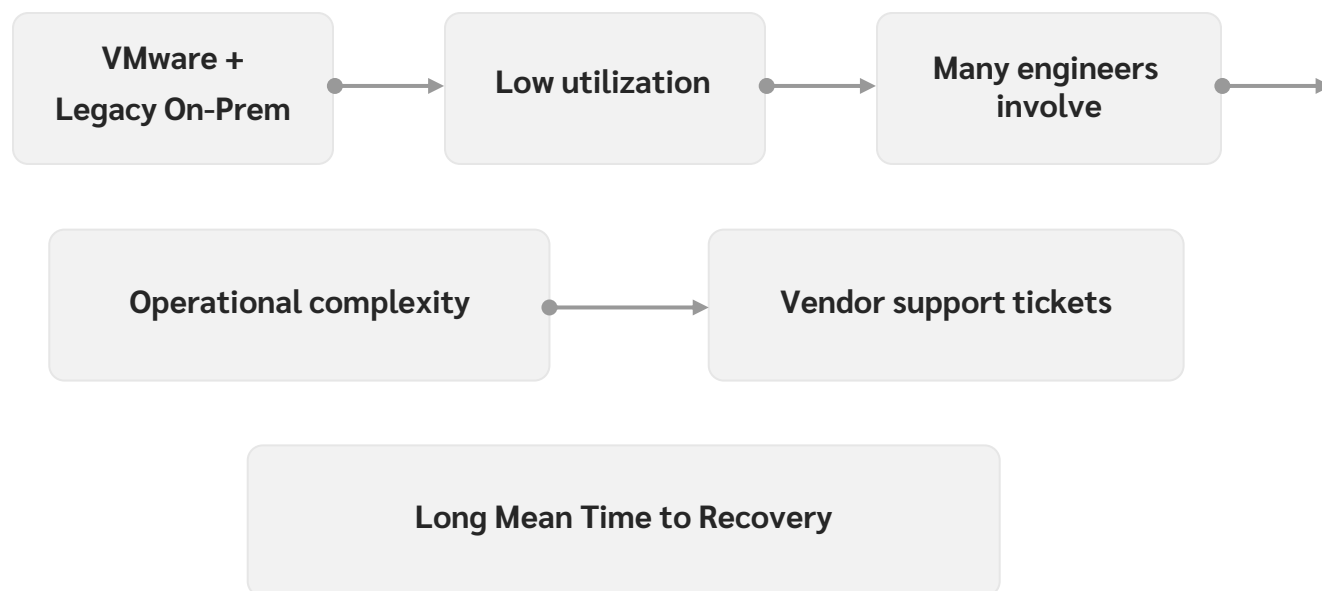
Key Takeaway

Five lessons for teams starting today

Our Legacy Infrastructure at That Time

2018 · STATE OF INFRASTRUCTURE

VMware + Legacy On-Prem → the hole we lived in



KEY POINT

The problem is not only the license.
Vendor trap existed in technology, operation, and incident response.

FRAGMENTED INFRASTRUCTURE

SERVERS

~**18,000** physical + virtualized
4 vendors · 10 overseas countries + Vietnam

STORAGE

~**110 SAN**
7 vendors

OPERATIONS

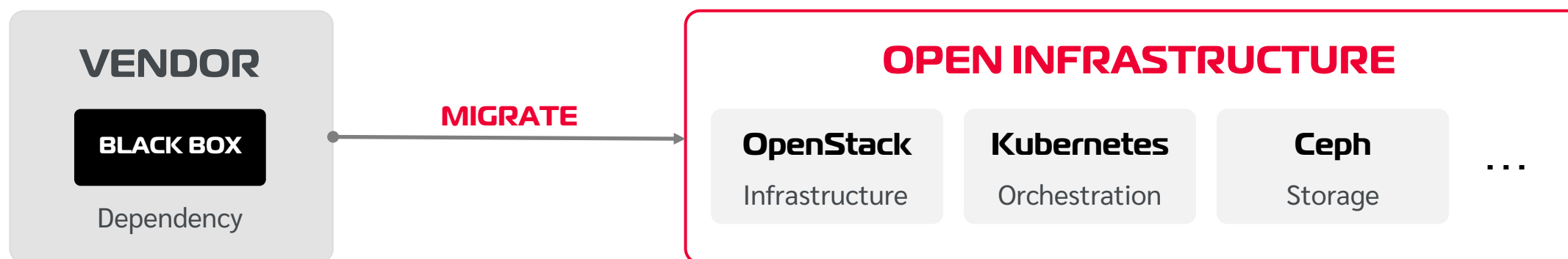
Manual deployment · Slow sizing
11 countries centrally operated
from Vietnam

THE TIPPING POINT

Scale ↑ — License bill ↑ — Operational dependency ↑

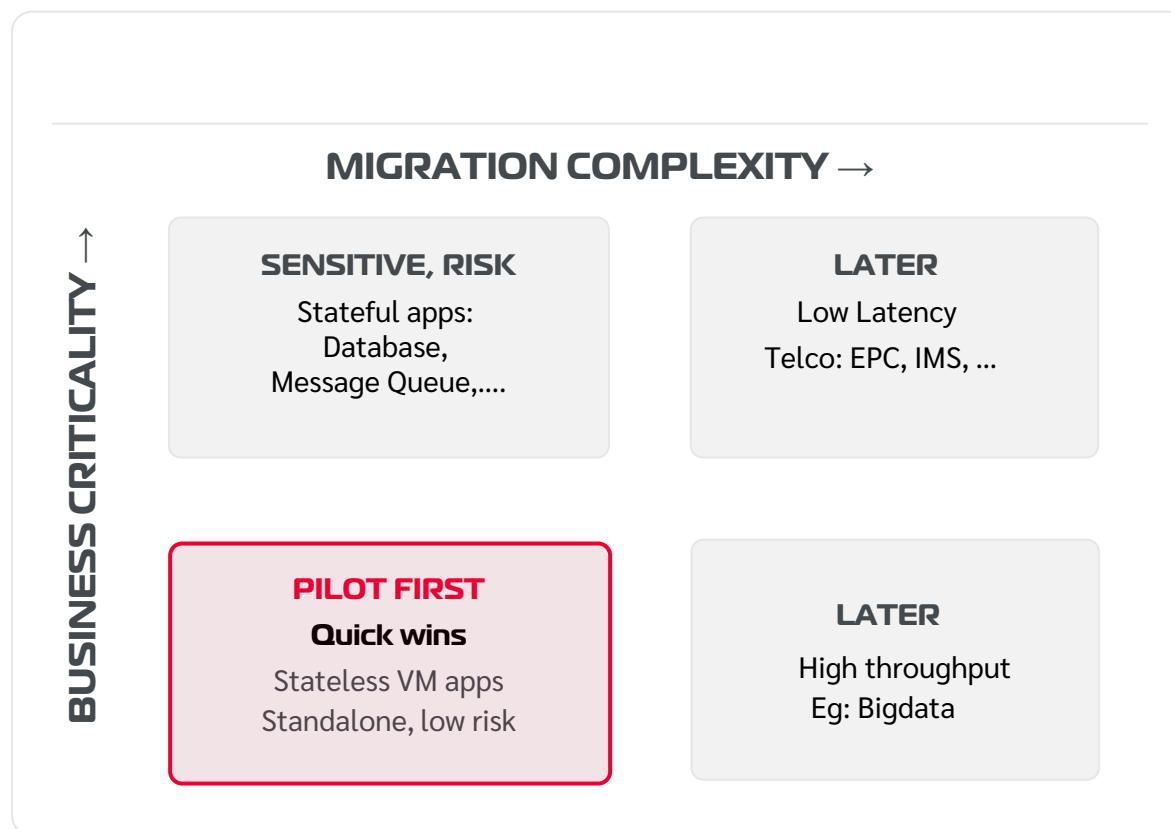
Strategy

From vendor dependency to technology ownership



Migration was never about replacing one product with another. It was about changing who controls our infrastructure.

Where Should We Start ?



FIRST WORKLOADS MIGRATED

NIMS	Nocpro	KPI Management	NPMS	
VSA	AAM	VMsa	VIPA	AOM

KPI FOR SUCCESSFUL PILOT

- Features work normally
- Performance: throughput & latency
- Communication with other apps normal

PRINCIPLE

We started where failure was survivable — and learning was valuable.

Challenges

FIRST MONTH IN PRODUCTION (2018): ~100 compute nodes + 5 rack-mounted storages

HUGE EFFORT

A large number of VM need to migrate

TEAM CAPACITY

<10 cloud engineers while >20 sub-project of Openstack, Ceph,

CONTROL CONNECTIVITY

How to control inbound and outbound connectivity after moving VM to cloud

AUTO-SCALING + AUTO-HEALING

OpenStack had no built-in autoscaling that integrated with Prometheus.

MONITORING

How many modules need monitoring

-> NEXT PHASE: upgrading ?

Steps for Migration

Each workload goes through the same steps — repeatable, automatable, measurable

01 · Gather spec, sizing

Total needed resources for running.

02 · Convert disks

App installation on new cluster.

03 · Creating VM

Creating VM (RAM, CPU, storage).

04 · Network Config

Network requirements (IP, ports, domain).

05 · Testing & Validation

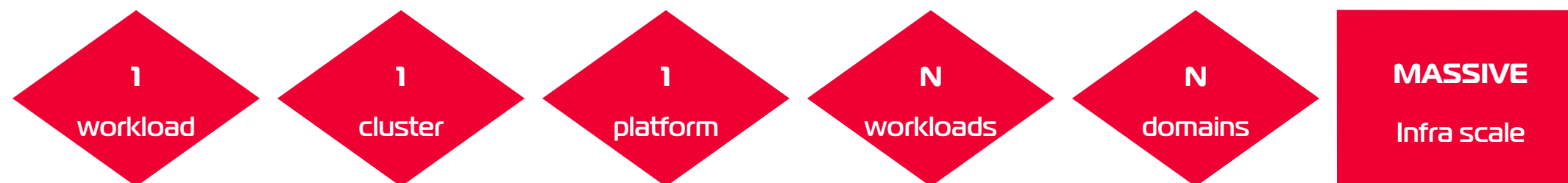
Run, test, compare with legacy.

06 · Monitoring & Backup

Monitoring integration + backup.

From a Pilot to a Platform

The first migration was not the victory. Repeatability was.



INSIGHT

The real breakthrough was turning migration from a single project into a factory.

Next phase: Kubernetes

PHASE A · INFRASTRUCTURE



OpenStack gives us the foundation — compute, storage, network as a programmable layer.

Compute

VM · baremetal

Storage

Block · object

Network

SDN · VXLAN

PHASE B · ORCHESTRATION



kubernetes

Kubernetes gives us the orchestration — workloads scheduled, scaled, healed automatically.

Apps

Customers

OSS/BSS

Operations/ Business

5G Core

Future telco

We didn't replace one vendor with another vendor

OUR OPEN INFRASTRUCTURE STACK

Layers built on community-driven open source

WORKLOADS

Applications

ORCHESTRATION

Kubernetes

INFRASTRUCTURE SOFTWARE

OpenStack · Ceph

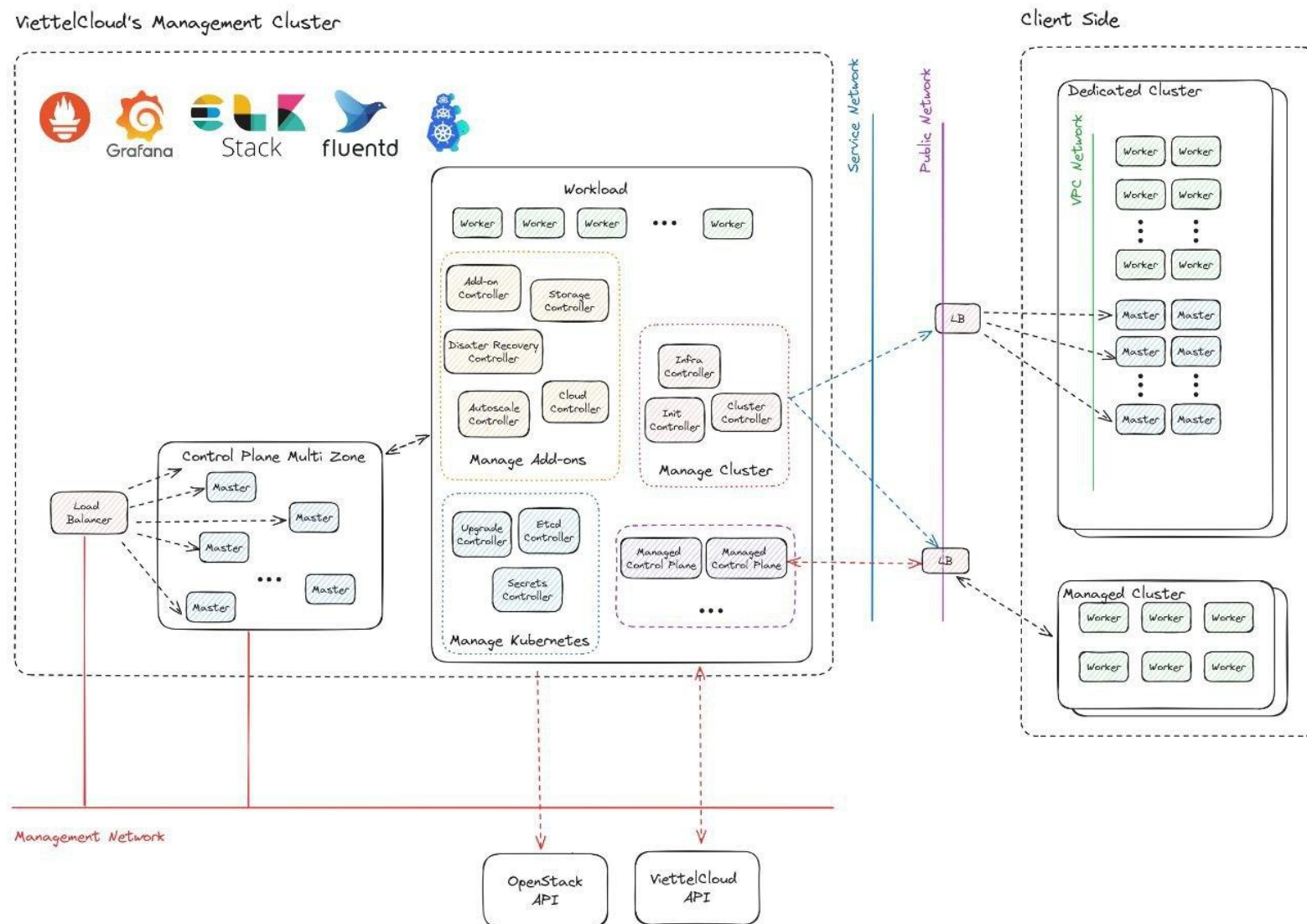
HARDWARE

Compute · Storage · Network
physical infrastructure

OBSERVABILITY

Prometheus · ELK

Viettel Kubernetes Engine



The Architecture Evolved With Us

4 DIMENSIONS · ONE PIPELINE

Our architecture was built step by step — technology, people, process, and organization — synchronized by a repeatable migration pipeline

FOUR DIMENSIONS

TECHNOLOGY

Cloud-native tools, automation, observability.

PEOPLE

Engineers reading source code, debugging upstream.

PROCESS

Standardize, automate, observe, feedback, repeat.

ORGANIZATION

Self-driven platform team, SRE culture.

MIGRATION PIPELINE

Assessment

understand workload

Target design

architecture fit

Migration

execute pipeline

Validation

production checks

Cutover

go-live

Observe

monitor, learn

Rollback / Improve

iterate

The Outcome (2018)

COMPUTE SCALE

128

compute nodes

3,800 cores · 20 TB RAM

APPLICATIONS

40

applications migrated

across business domains

KUBERNETES FOOTPRINT

1.3K

vCPUs

2.4 TB RAM · 13 TB storage

production K8s at scale

LICENSE COST SAVED

Millions of USD

PROVISIONING TIME

~~A few days~~

Physical manual process

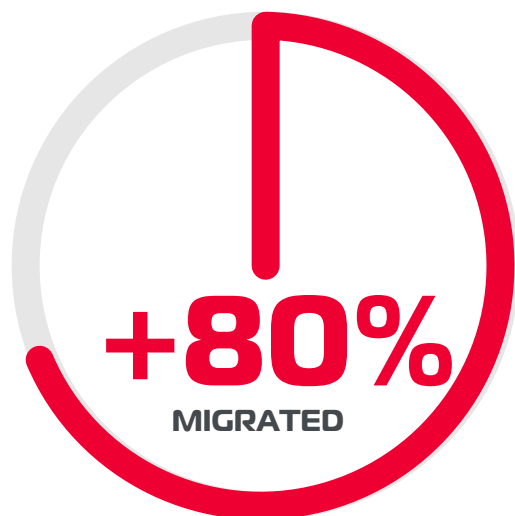


A few minutes

automated provisioning

Today: +80%+ on Open Architecture

Three outcomes — Three sovereignties



3 BENEFITS

Resource Efficiency

Higher utilization, lower idle capacity.

Cost Optimization

License savings, predictable TCO.

Technology Ownership

We own the roadmap, not the vendor.

3 SOVEREIGNTIES

Data Sovereignty

Local data, local control.

Operational Sovereignty

Internal expertise, internal runbooks.

Technology Sovereignty

Upstream contributions, vendor freedom.

Scale

CLOUD INFRASTRUCTURE

13**Data Centers**

Vietnam + overseas countries

~17K**Server Racks**

deployed capacity

~5K**Compute Nodes**

private + public cloud

~30K**Virtual Machines**

running workloads

KUBERNETES

~200**Clusters**

across business units

~2K**K8s Nodes**

worker nodes total

~20K**vCPUs**

scheduled capacity

~40K**GB RAM**

memory total

BUSINESS REACH

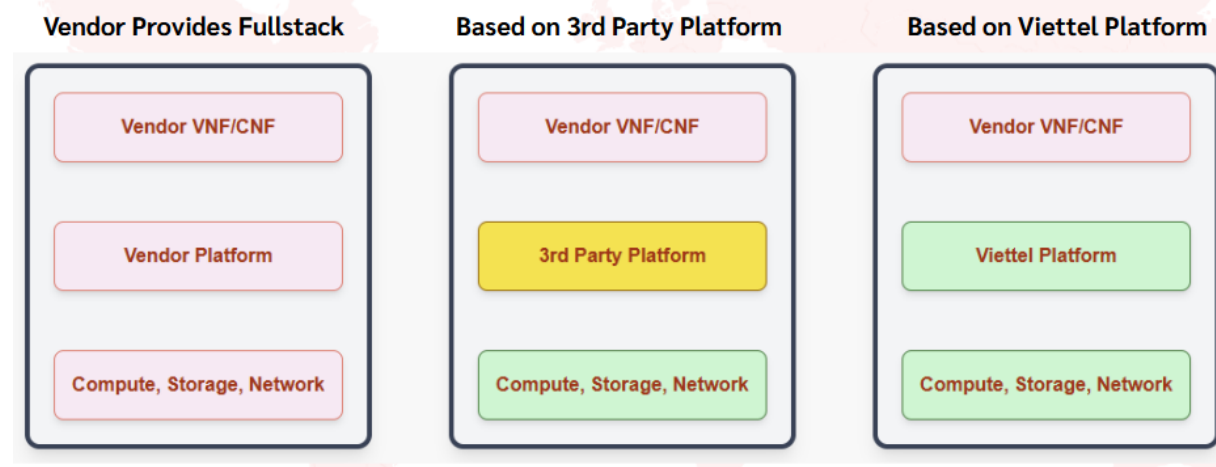
Domestic: Public Cloud · Ministries / Provinces · Telecom operators**Overseas:** Peru · Myanmar · Burundi · Tanzania · Cambodia · Timor-Leste · Laos

Not Only IT Cloud, but Also Telco Cloud

WHY TELCO OPERATORS NEED CLOUD

Cloud will become the operating model for the next-generation telco

- 1 **Faster Deployment**
1-2 months → 1-2 weeks for new services.
- 2 **CAPEX Optimization**
Move from upfront hardware spend to elastic capacity.
- 3 **Lower TCO**
~50% reduction over a 5-year horizon.
- 4 **OPEX Reduction**
Automate routine ops; engineers focus on innovation.
- 5 **Agility & Flexibility**
Spin up workloads for testing, scaling, new business models



We didn't eliminate operations. We changed the work.

WHAT CHANGED FOR ENGINEERS

Open source did not reduce work — it shifted the engineer's focus from firefighting to contribution

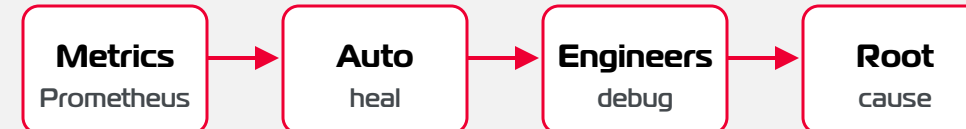
BEFORE — LEGACY MODEL



EXPERIENCE

Long wait, no insight, every fix routed through vendor.
Engineer felt like a dispatcher, not a builder.

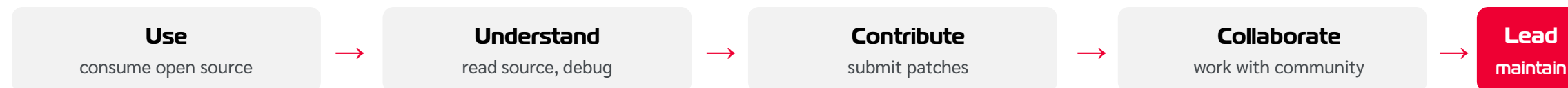
AFTER — OPEN ARCHITECTURE



EXPERIENCE

Engineer reads source code, understands root cause, contributes fix upstream. Builds, not dispatches.

THE OPEN-SOURCE JOURNEY



Community Became Part of the Migration Strategy

WE COULD NOT SCALE ALONE

“When you remove a proprietary vendor, you need a different network of expertise.”

- National Network: Collaborated with local universities, local engineers, and tech startups to foster a robust systems engineering community in Vietnam.
- Regional Network: Shared best practices and collaborated across Asian markets, speeding up regional engineering knowledge cycles.
- Global Integration: contribute into OpenInfra, Cloud Native Computing Foundation (CNCF), and upstream community groups.

FROM CONSUMER TO BOARD MEMBER

Active contributor is the final maturity of escaping dependency.

 **LEADERSHIP:**

Platinum seat on the Board of OpenInfra Foundation

 **CONTRIBUTOR:**

Submitting bug fixes, core code patches upstream

 **OPERATOR:**

Running complex scale, compiling custom packages

 **CONSUMER:**

Using open packages out of the box

Lessons From the Journey

Five lessons for teams starting their migration journey today

1

Don't migrate everything at once.

Find the right starting point — where failure is survivable and learning is valuable.

2

Don't optimize for technology replacement.

Optimize for operational ownership — who runs the system when something breaks.

3

Build the migration factory early.

Repeatability beats heroics — make every workload follow the same eight steps.

4

Invest in people as aggressively as infrastructure.

Open source requires internal expertise — engineers must read source code, debug upstream.

5

Join the community before you need the community.

Upstream collaboration accelerates scale — and ultimately gives you a seat at the table.

THE REAL TRANSFORMATION

We migrated ownership

OWNERSHIP OF

Technology

upstream contributions

Operations

internal SRE culture

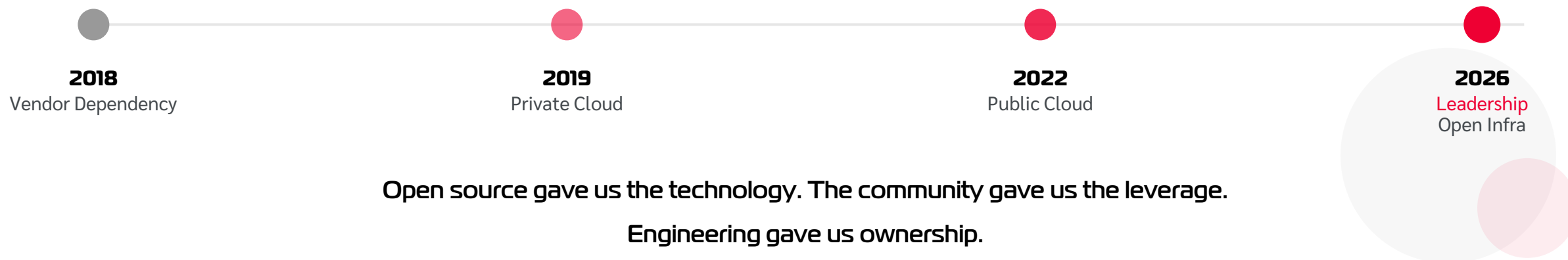
Knowledge

source-level expertise

Roadmap

our decisions, not theirs

JOURNEY · 2018 → 2026



THANK YOU

KubeCon + CloudNativeCon + OpenInfra Summit + PyTorch
Conference China 2026

VIETTEL CLOUD ECOSYSTEM powered by OpenStack

PaaS

Containers



Hosting



Cloud Security



Database



Interconnection



IOT Platform



AI & Machine Learning



MSP

Cloud Security



Professional Services



SaaS

Surveillance Camera Service



Cloud Security



Hosting



Cloud Applications



IaaS

Cloud computing



Network Virtualization



Load Balancing



Data storage



Data backup



Interconnection



DATA CENTER

Data center service



Professional Services

